Propaganda Detection Model

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New Geography of Information Age

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9th December 2024

Introduction

Propaganda has been prevalent for many decades. The word was first used in the 1500s for the spreading of Protestantism. It has been identified as an issue throughout history. Political propaganda is defined as information of a biased or misleading nature which is used to promote or publicise a particular political cause or point of view (Jowett & O'Donnell, 2012). It uses emotional appeals, half-truths, and manipulation to influence public perception, making it a useful tool in shaping ideologies, behaviors, and societal norms (Sprenkamp et al., 2023). In the digital age, its reach and abilities have grown a lot, raising significant ethical and societal concerns.

Most importantly, we need to look at its impact at the grassroots level. Grassroots stakeholders are the people who are consuming this propaganda content. They are usually the target of these misinformation campaigns and hold the power to make important decisions that the propagandists need to be influenced. These campaigns can be used to manipulate voting behaviors and directly affect democratic processes as well as societal cohesion (Miller, 2018).

With large language models (LLMs) such as GPT-3 and GPT-4 the creation and distribution of propaganda has become ever easier. These models can be used to craft convincing and highly targeted propaganda messages at a large scale. For example, they can generate narratives which are tailored to exploit psychological biases, such as fear or group identity, with minimal human oversight (Zellers et al., 2019). The advancement in these LLMs allows them to mimic credible sources, making it more difficult to differentiate authentic information from propaganda. These models can also replicate rhetorical patterns that increase the credibility and virality of such content, making them the ideal tools for people looking to spread propaganda (Rashkin et al., 2020).

This problem is more severe in online media, where platforms like Twitter and Facebook amplify these messages through algorithms that prioritise engagement over honest information (Vosoughi, Roy, & Aral, 2018). The reach of propaganda in such spaces increases its harmful effects, leading to societal polarisation, erosion of trust, and even real-world violence (Starbird et al., 2020). This shows an urgent need for proper propaganda detection mechanisms.

In this project, I explore the harm caused by propaganda, and the broader implications for grassroots stakeholders and society at large. By addressing these challenges, I aim to contribute to the development of scalable, ethical, and accessible solutions for detecting and mitigating the impact of propaganda in the digital age.

Research Hypothesis : Messages exhibiting higher emotional intensity, frequent use of rhetorical devices, and greater inclusion of profanity are more likely to be classified as propaganda than non-propaganda.

Literature review

The concept of propaganda has evolved a lot since it was first introduced in the 1500s in terms of religion. The term was formally introduced in 1929 specifically in reference to war propaganda (Fellows, 1959) which included a lot of articles dealing with fascist, Communist, or Nazi propaganda. Whereas, the current definition plays emphasis on the political aspect of propaganda with psychological features such as cognitive dissonance, heuristics and so on to influence public opinion (Pratkanis & Aronson, 2001).

With the rise in different forms of media, we can also see a rise in the ways propaganda can be spread. Online or digital media has played a huge role in the spread of digital propaganda. Studies suggest that propaganda now thrives on an "attention economy," where the algorithms tend to prioritise sensationalization and polarisation of content to increase user engagement (Tufecki, 2017).

Moreover, anonymity or fake profiles over the internet take away the accountability of sources of information. Online media also makes it much easier for the propaganda to reach a larger number of people. Additionally, Large language models have made it even easier to produce propaganda messages on a large scale such that they can also mimic credible sources. This shift has made it even more important to be able to detect propaganda since its impact has severe societal consequences (McNamee & Parakilas, 2018). Zellers (2019) demonstrated how GPT-2 could produce convincing fake news articles, raising ethical concerns about the misuse of AI. Moreover, studies indicate that LLMs can reinforce biases present in their training data, which may align with propagandistic narratives (Bender et al., 2021).

As much as LLMs are harmful, they are also helpful in the detection of propaganda. A lot of computational propaganda detection approaches have been recently developed. Early methods

employed basic natural language processing (NLP) techniques, such as keyword matching and sentiment analysis (Bruckman et al., 2018). Recent advancements use machine learning models to analyse complex linguistic features including context, tone and semantic relationships. Techniques like BERT (Bidirectional Encoder Representations from Transformers) and GPT-3 have demonstrated high accuracy in text classification tasks, including distinguishing propaganda from neutral content (Devlin et al., 2019). However, these models sometimes still struggle with subtle propaganda (Rashkin et al., 2017). Few of the popular propaganda detection models include Proppy and PROTECT. Proppy is a news-based propaganda detection model designed to identify propaganda in written articles. It uses natural language processing (NLP) techniques to extract linguistic features such as bias, sensationalism, and subjectivity. Proppy classifies text as propaganda or non-propaganda by analysing headline patterns, word frequencies, and sentence structures. On the other hand, PROTECT (Propaganda Detection with Textual Evidence and Contextual Transparency) helps in detecting propaganda on social media. It uses contextual embedding (a technique that captures the nuanced meanings of words based on their surrounding context) and multimodal analysis (which means it considers other forms of media (like images or videos) along with textual content) to understand the textual as well as contextual analysis. This allows the model to better interpret ambiguous language, such as sarcasm, idioms, or emotionally charged statements, which are often used in propaganda.. It uses attention mechanisms to focus on particular parts of the text which later aids in classification.

Even though there are quite a few models which help in detection of propaganda, we can’t really see them being implemented on platforms. Recently, we can see a lot of propaganda is spread through whatsapp and twitter during elections. This targeted propaganda during elections in multilingual regions, such as India and Kenya, often uses culturally specific symbols

and idioms to manipulate public opinion (Bradshaw & Howard, 2017). For prevention, The government established a fact-checking unit under the IT Ministry to verify news and identify misinformation circulating online. Existing legal frameworks such as the Information Technology (IT) Act have been used to address misinformation and propaganda. Bill No. 47 passed in Lok Sabha in 2023 addressed Fake news on Social Media. The bill suggests that any individual on social media found guilty of posting fake news may face imprisonment of up to seven years and/or a fine of up to ₹10 lakh. Additionally, If a company is found guilty, all responsible individuals in charge at the time of the violation, such as directors or managers, may be held accountable unless they can prove they had no knowledge of the contravention or took due diligence to prevent it. Amendments in intermediary guidelines require social media platforms to appoint compliance officers, remove flagged content within a specified timeframe, and maintain traceability of harmful information. Platforms like Twitter, WhatsApp, and Facebook have been asked to improve their mechanisms for identifying and curbing fake news. WhatsApp, for instance, was required to limit message forwards and add fact-checking tools in response to issues like mob violence caused by false information.

For implementation of this propaganda detection model, there are several features that seem to be important. Style, vocabulary richness, and readability - prefixes, suffixes, and punctuation marks, type-token ratio, Representations based on writing style and readability, panning readability and vocabulary richness seem to be quite important as per Maarouf (2023).

Propaganda detection model Proppy suggests that Name Calling ((e.g., “radicals,” “traitors”), Glittering Generalities, virtue-laden words (e.g., “freedom,” “honor”), Linking the authority or prestige of a respected symbol (e.g., a national flag) to an unrelated concept, Relying on endorsements from authority figures, celebrities, or experts to lend credibility to a message,

portraying propagandists as relatable, ordinary people to build trust, Manipulating information by omitting, exaggerating, or misrepresenting facts to support a particular narrative, encouraging conformity by asserting that “everyone is doing it” or “the majority agrees” are some of common methods used in propaganda messages (Cedeno et al., 2019). PROTECT (Vorakitphan et al., 2022) similarly talked about other features such as Reiterating the same message to instill it in the audience’s mind, Using short, memorable phrases to encapsulate an idea (e.g., “Make America Great Again”), Instilling fear of alternatives to support a particular idea, Overstating or downplaying elements to manipulate perceptions, Appealing to patriotism or group identity, Using emotionally charged or stereotypical terms, Dismissing ideas by associating them with groups hated by the audience, Attributing complex phenomena to a single, simplistic cause, Using ambiguous language to confuse or mislead, Citing authority figures as evidence without additional justification, Presenting only two options when many exist, Using phrases to shut down debate (e.g., “It’s just the way it is”), Distracting from the main issue by introducing irrelevant information and Misrepresenting an argument to easily refute it.

Methods

I implemented machine learning to make a model that would help me understand which features are the most important in identifying propaganda from non-propaganda. For the dataset, I reached out to professor Maarouf, Abdurahman at LMU Munich because of his published paper about the high quality dataset for propaganda who kindly shared his dataset with me.

I used the HPQ+ dataset, which contains around 30,000 labeled tweets categorised as propaganda or non-propaganda. The dataset was pre-labeled and preprocessed, making it easier for analysing propaganda detection. The tweets were collected using hashtags and keywords related to topics like elections and social movements, covering all important events from 2018 to 2022. This ensured the data was not only relevant to the study but also geographically diverse.

To identify propaganda in tweets, I extracted several features based on my literature review. First, I used sentiment analysis tools like TextBlob to measure polarity and subjectivity, as propaganda often relies on emotionally charged or opinion-based language. Next, I evaluated profanity using a curated list of words with severity ratings, as provocative language can be a sign of manipulative content. I also analysed text length metrics, such as word count and average word length, to see how message structure could mean propaganda. I also developed a custom method to detect rhetorical devices like exaggeration, name-calling, and slogans, which are common in persuasive or misleading content. This was based heavily on proppy's method of propaganda detection.

For the model, I used a subset of the dataset, with 20% selected for training and testing. This subset was split into 80% for training and 20% for testing to balance computational efficiency and evaluation. I then employed TF-IDF vectorization to quantify the textual data, focusing on unigrams and bigrams while filtering out common stopwords. Using logistic

regression, I trained the model to classify tweets as propaganda or non-propaganda. Logistic regression was chosen for its simplicity and interpretability, which allowed me to examine the importance of different features.

The model's performance was assessed using precision, recall, F1-score, and a confusion matrix. To better understand how different features contributed to the classification, I analysed the coefficients from logistic regression and visualised the results in the form of a bar chart.

However, there were several limitations as well. The predefined lexicons and rhetorical patterns might have overlooked context-specific propaganda strategies. Additionally, TF-IDF and n-gram approaches, though effective, do not capture deeper semantic relationships in text. Using a smaller subset of the data for training and testing may also have constrained the model's generalisability.

Alongside this, I also conducted a survey to explore why such detection methods are not widely implemented at the grassroots level. The survey's main purpose was to identify how to make propaganda detection convenient for the content consumers. The form was distributed using snowball as well as purposive sampling to make sure people mainly of demographic 35-70 were reached since they consume the most amount of news content. I believe this qualitative input is important for bridging the gap between academic solutions and practical applications in combating propaganda.

Results

Using the code, we trained the model to differentiate propaganda from non-propaganda and moreover also identified the most important features. The model was able to classify propaganda from non-propaganda with 87% accuracy. The model performs well in identifying non-propaganda content, with a high precision of 0.87, recall of 0.93, and an F1-score of 0.99. The model performs okay in identifying propaganda content, with a precision of 0.72, recall of 0.18, and an F1-score of 0.29.

Among the features, false dilemma count was the most important, with a coefficient of 0.551939. Bandwagon count and slogan count display negative coefficients of -0.501730 and -0.475108, respectively. Subjectivity, Profanity count, Appeals to authority and Emotion were moderately important with coefficients of -0.319652, 0.310287, -0.259138 and -0.236247 respectively. Polarity (-0.184548) and Profanity severity (-0.177150) have lower but still significant importance. Less impactful features include word count (0.024363) and character count (-0.000405).

The survey helped me understand the feasibility and employability of a propaganda detection tool and following were the results :

- Age Demographics
    - The majority of respondents were aged 35–50 (38.5%) and 18–24 (32.7%).
    - Smaller proportions were from the 50–70 group (19.2%), while under 18 and 70+ groups had minimal representation (1.9% each).
- Occupation
    - Professionals dominated (55.8%), followed by students (30.8%).
    - Other occupations accounted for 11.5%, and retired individuals were 1.9%.

- Consumption of News
    - A majority reported consuming news daily (often) (53.8%), with 28.8% consuming it sometimes (3–5 times a week) and 17.3% rarely (1–2 times a week).
- Concern for Propaganda
    - Somewhat concerned respondents were the largest group (42.3%), followed by very concerned (25%).
    - Neutral responses accounted for 13.5%, while a smaller group was not very concerned (17.3%) or not at all concerned (1.9%).
- Most participants (59.6%) reported encountering content they believed to be propaganda, while 17.3% had not, and 23.1% were unsure.
- Interest in Propaganda detection tool
    - A significant portion expressed interest (Yes: 59.6%), with fewer respondents indicating Maybe (28.8%) or No (11.5%).
- Propaganda detection tool as a browser extension
    - Most respondents were somewhat likely (52%) or very likely (32.7%) to install a browser extension for propaganda detection.
    - A smaller group indicated they were unlikely (15.4%) or very unlikely (1.9%).
- Propaganda detection tool as a screenshot uploading
    - Somewhat likely (30.8%) and very likely (26.9%) responses dominated, while somewhat unlikely (23.1%) and very unlikely (19.2%) were also noted.
- Privacy Concerns: The majority (76.5%) cited this as a major deterrent, with concerns including personal information leakage and a general reluctance to grant online tools access to personal data.

- Distrust of Accuracy: A significant proportion (41.2%) doubted the tool's reliability in identifying propaganda correctly.
- Lack of Need: Nearly a quarter (23.5%) felt no need for such a tool in their online habits.
- Fear of Censorship: A smaller group (7.8%) expressed concerns about the potential for censorship or restriction of free speech.

Discussion

Based on the literature review and survey results, deploying a propaganda detection model as a browser extension may seem convenient, but it raises significant privacy concerns. Users are wary of extensions that could potentially read all data displayed on their screens, as this infringes on their personal privacy. On the other hand, using a tool that requires users to take a screenshot and upload it for analysis appears cumbersome and impractical. This approach also assumes that users are critically evaluating the content they consume and are aware enough to suspect propaganda, which may not always be the case.

A feasible alternative, particularly for combating propaganda on platforms like WhatsApp, could be a bot designed for this purpose. Users could forward questionable messages to the bot, which would then analyze and identify whether the content is propaganda. This method addresses privacy concerns by avoiding continuous monitoring and does not require installing an extension. However, it still relies on users' awareness and willingness to forward messages for evaluation. Without a critical mindset, users may not engage with the tool effectively.

If privacy concerns were set aside—acknowledging that many platforms already violate privacy by collecting extensive personal data—the question arises: why have these platforms, despite their vast resources, not implemented propaganda detection models? The reluctance of platforms to counter propaganda can be explained by several factors.

First, combating propaganda conflicts with their business models. Platforms like Facebook and WhatsApp generate revenue from engagement, and propaganda often drives high levels of interaction, such as shares, comments, and views. If propaganda dissemination is

restricted, platforms might lose revenue from advertisers and propagandists who exploit these tools.

Second, platforms may have political affiliations that create conflicts of interest. For example, reports have suggested that Facebook has ties to political parties such as India's Bharatiya Janata Party (BJP). An Article-14 investigation revealed Facebook's preferential treatment towards India's ruling Bharatiya Janata Party (BJP), despite enforcing its rules against political bias more strictly on the opposition Congress Party. Before the 2019 elections, Facebook removed 687 Congress-linked accounts for undisclosed affiliations but took no comparable action against BJP-linked accounts, despite their significant presence and undisclosed links to the party. Facebook has also partnered extensively with the Indian government on projects across education, tribal welfare, and public health, leading to criticism of a conflict of interest and the company's dependence on these collaborations for audience expansion and revenue. Allegations against Facebook include ignoring hate speech by BJP members and benefiting from commercial ties with the Modi administration.

However, WhatsApp seems to have taken proactive steps to combat misinformation by partnering with International Fact-Checking Network (IFCN)-certified organizations. Since 2018, over 50 fact-checking organizations worldwide have collaborated with WhatsApp using tools like the WhatsApp Business App and Business Platform. These partnerships aim to connect users with reliable and accurate information, particularly through user-initiated reporting.

If a user encounters suspicious or potentially false information, they can forward it to a trusted fact-checking organization in their region. These organizations then review the content and respond with relevant fact-checking articles. This method ensures user privacy since

WhatsApp messages are end-to-end encrypted, making them accessible only to the sender and receiver.

In addition to ongoing partnerships, WhatsApp has invested significantly in strengthening the fact-checking ecosystem. Recent initiatives include awarding grants to support efforts combating AI-generated misinformation and launching a "Deepfake Hotline" in India ahead of the 2024 elections. Such measures aim to address both the challenges posed by technological advancements and the specific needs of countries during sensitive periods like elections. However, the platform's reliance on user-initiated reporting underscores the need for user awareness and active engagement in combating propaganda and fake news. Integrating features like these into tools developed for broader propaganda detection could enhance their effectiveness and public trust.

Twitter seems to also be taking steps, where they have launched a new policy in 2022 where they will add warning labels on misleading information related to crisis (war or natural disasters). Twitter said it will require verification from multiple credible and publicly available sources to determine what is misinformation and what is not (The Economic Times, 2021)

In 2017, Facebook wrote about their ways of combating misinformation. They said they will be disrupting economic incentives for spreading false news, developing tools to identify and reduce its spread, and promoting informed decision-making. The company leverages machine learning to detect inauthentic accounts, enforces ad policies against spammers, and collaborates with third-party fact-checkers to flag misleading stories. Meta also improves News Feed rankings to de-prioritise false content and supports public awareness initiatives like the Facebook Journalism Project and the News Integrity Initiative.

If we were to implement these models as an extension or a bot, another important thing is how would we market it so that a large number of people reach it. Most importantly, due to the third person effect, people tend to believe other people are more likely to be influenced by media messages. Hence, I believe that marketing needs to be done in that way as well. It's important that we focus on the aspect of saying that you are helping other people around you not fall prey to propaganda instead of yourself. Making them feel like righteous citizens helping those around them are more likely to prompt them into action. Especially the generation we are targeting would be the demographic of 35+ and based on socioemotional selectivity theory older people prioritise emotional connections and benefit from roles where they can share knowledge, thus feeling like valued contributors or mentors.

Fieldwork

To understand the conversation around the topic I reached out to the following experts:

1. Professor Anirban Sen who has extensively worked on propaganda in terms of network analysis. These are the notes from my interview with him:

- Based on your research, what are the key strategies used in online political propaganda to influence public opinion?
    - Long cascades
    - Incendiary speech
    - CSCW paper - long tweets vs short tweets (propaganda vs non propaganda)
    - Profanities
- How have these strategies evolved with advancements in computational tools?
    - Tweets generated through LLMs (hate speech)
    - Hate speech standard
    - Easier to write content but network dynamics are the same
- How do data privacy violations facilitate these political propaganda?
    - Geotagging api
- I read about a model called PROTECT - which uses NLP specifically BERT pre trained models to identify propaganda messages, are there any other common or better ones out there, and why do you think they haven't been implemented.
    - Proppy
    - Policy regulation on platforms(twitters)
    - Several levels of bureaucracy

    - Countering propaganda is against their business models. Platform loses revenue if propagandist don't spread messages

Other useful notes

- Look at retweet tags and network dynamics to see if same users are present in your database
- Crawl your own whatsapp groups
- Look into MCA - misinformation combat alliance
- Dataset include snapshots of data and may not hold true for general data in real world
- Linguistic features of tweets correlating with network dynamics

2. Email conversation with Miss Sushma Gaur who was a professional at DAVP (Directorate of Advertising and Visual Publicity) and is currently working as PR manager for the Ministry of Railways :

- How do you ensure clarity and relatability in messages intended for diverse audiences across regions?
    - In today's interconnected world, effective communication across regions has become crucial. Building trust and connection with individuals from diverse backgrounds is essential for successful cross-cultural interactions. To navigate these situations successfully, it's important to develop strong interpersonal skills that foster understanding and create meaningful connections.
    - Some of the key strategies involves:

1. Know your Audience: To communicate effectively, you must first understand who you're talking to. This means researching your audience's demographics, cultural backgrounds, interests, and preferences. By doing so, you can anticipate their needs and expectations, and tailor your message accordingly

2. Simplify Language: Use clear and simple language. Focus on direct and straightforward language that can be easily translated or interpreted across cultures. Slang and idioms to be avoided

3. Make sure your message is adapted (not just translated) to fit the language and culture of each region.

4. Pictures and symbols can help your message reach people, even if they don’t speak the same language. Be careful with colors and images because they can mean different things in different cultures. By keeping things simple, clear, and sensitive to cultural differences, your message will be more likely to connect with people everywhere.

- How do you determine the most effective mediums (e.g., print, digital, outdoor advertising) for reaching diverse demographics?
    - To determine the most effective mediums for reaching diverse demographics, we need to consider several factors that can vary by region, age, income, interests, and technology usage.

- It is important to consider several factors like your target audience, budget, goals, and timing. For example, if you are targeting a younger audience, digital format might be the best fit for you. If your target audience is an older demographic, traditional tools could have better results. Start by thinking about who your target audience is and what your goals are.
- Digital channels like social media, email, and websites are great for targeting specific groups, especially in younger or urban populations. TV, radio, and print (newspapers, magazines) can be more effective for reaching older demographics or those in less digitally connected areas. Billboards, transit ads, and posters work well in urban or high-traffic areas. They are great for broad awareness campaigns and can be effective for certain demographics, especially when you want to create a visual impact.
- Use of analytics tools helps to monitor how different mediums perform with the audience. Digital platforms, in particular, offer a wealth of data on user engagement.
- By understanding the audience's behavior, preferences, and where they spend their time, we can choose the most effective mediums for messaging.

- How do you measure the impact of a public awareness campaign in terms of reach and effectiveness?
  - Measuring the impact of a public awareness campaign involves assessing both reach and effectiveness of the campaign. Following steps can help you develop a solid approach for how to measure a campaign effectiveness:

a) Measure Reach: Track how many people saw your campaign (through social media views, TV broadcasts, print circulation, or outdoor reach). Check if your campaign reached the right age group, gender, or target audience.

b) Measure Effectiveness: Ask the audience before and after the campaign if they're more aware of the issue or have changed their behavior. Measure the number of social media interactions, website visits, and click-through rates on digital formats. Monitor press coverage and online mentions to see how much buzz your campaign generated.

c) Evaluate whether the campaign's results in reach, engagement or behavior change.

- What role does public feedback play in shaping the direction of such large-scale campaigns?
    - Public feedback plays a crucial role in shaping the direction of large-scale campaigns by helping to refine strategies, improve messaging, and ensure the campaign resonates with the target audience. It reveals whether the campaign is resonating with the audience, highlights areas that need adjustment, and helps manage any negative reactions.
    - By gathering insights from feedback channels such as surveys, reviews, and social media, we can identify patterns and trends, enabling them to tailor the strategies to better communication.
    - By analyzing feedback, we can identify areas for enhancement.

- Listening to feedback also boosts engagement, as people feel heard and involved, and provides valuable insights for future campaigns.

- How do you address potential misconceptions or resistance through messaging strategies?
  - No one understands your work quite like you do. What we think as experts is universally understood – it's not.
  - To address misconceptions or resistance in messaging strategies, key steps to be followed:

    a) Start by recognizing the audience's doubts or misconceptions. This shows empathy and builds trust.

    b) Correct misinformation with clear, simple facts. Use trusted sources and data to back up your message. Make it easy to understand and relatable.

    c) Reframe negative perceptions by focusing on positive outcomes or benefits.

    d) Share real-life examples, stories, or testimonials from people who have had positive experiences. Personal stories can humanize the message and make it more relatable.

    e) Encourage feedback and create spaces for conversation. Show that you are listening and open to discussion, whether through social media, Q&A sessions, or community forums.

f) Align your message with the audience's core values and beliefs. If people feel that your message aligns with what they care about, they are more likely to embrace it.

- What ethical considerations do you believe are essential when using public data for targeted messaging, especially with regards to data privacy and consent?
    - When using public data for targeted messaging, several ethical considerations are crucial, particularly around data privacy and consent which includes:

        a) Ensuring that individuals are fully aware of how their data will be used. This includes informing them about what data is collected, how it will be used, and the potential for targeted messaging.

        b) Be transparent about your data collection practices. Let users know what data is being collected, why it's being used, and who will have access to it.

        c) Minimum data for the purpose to be collected. Regularly review and delete outdated or unnecessary data.

        d) Be mindful of how your messaging impacts different demographics and ensure that it is inclusive and respectful.

        e) Avoid manipulation, ensure compliance with legal standards, and protect vulnerable populations.

- How do you ensure transparency and accuracy in public messaging while also maintaining a positive narrative.
    - Ensuring transparency and accuracy in public messaging while maintaining a positive narrative involves balancing honesty with optimism. The foundation of any effective message is factual accuracy, so it's essential to present clear, evidence-based information that is easy to understand without oversimplifying or distorting the facts.
    - Acknowledging difficulties shows transparency, but highlighting the positive actions being taken or the steps towards improvement keeps the message hopeful. Transparency also involves being open about limitations—avoiding overpromises and setting realistic expectations about what can be achieved, especially when the situation is evolving. Engaging the audience by encouraging feedback, asking questions, and providing regular updates ensures continued communication and reinforces trust. In this way, a message can remain both truthful and positive, reassuring the audience that they are being informed and involved in a meaningful way.

- What key considerations are important when crafting public awareness messages for large-scale projects?
    - When crafting public awareness messages for large-scale projects, it's essential to consider several factors to ensure the message is clear, effective, and engaging. Understanding the audience is crucial—tailoring the message to their demographics, values, and cultural context helps it resonate. The message should

be simple and concise, free from jargon, and focused on key points to ensure accessibility.

- Messaging should be tailored to the audience and should answer the questions that they may have.
- It is also crucial to consider the medium through which the message will be delivered, as this can have a significant impact on how the message is received. The communication should also be backed up by facts and evidence where appropriate, to build credibility and establish trust with the audience.
- The needs and wants of the target audience play a significant role in crafting effective messages that resonate with them. An effective way to identify the needs and wants of the audience is to engage with them directly. By actively listening to what the audience has to say, we can identify areas where you can improve your messaging and better meet their needs and wants.

Limitations

1. The results suggest a high number of false negatives, this may be due class imbalance, that is a higher number of non-propaganda texts than propaganda text. Taking into account class balance while training might give us a better model.
2. In depth exploration of technical working of the preventive mechanisms employed by platforms like facebook, whatsapp and twitter might be helpful in understanding the lacking better.
3. The lack of random sampling for the survey may introduce bias in the findings.

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